Wireshark and Network Models

1. Capture Flipkart Traffic Using Wireshark

Steps to capture packets

1. Open Wireshark on your computer.
2. Select the active Wi-Fi or Ethernet network interface.
3. Click Start Capturing Packets.
4. Open your browser and visit the Flipkart homepage.
5. Refresh the page using Ctrl + R.
6. Return to Wireshark and stop the capture.
7. Use these display filters to examine different protocol headers.

TCP/IP Layer	**Wireshark Display Filter**	**What to Look For**
Application	http or dns	HTTP requests or DNS queries, if present
Transport	tcp or udp	Source port, destination port, sequence number
Internet	ip or ipv6	Source and destination IP addresses
Network Access	eth or wlan	Source and destination MAC addresses, where visible

Important: Flipkart generally uses HTTPS, so the HTTP content may be encrypted. You may see TLS records, TCP segments, DNS queries or QUIC traffic rather than readable HTTP messages. Also, Wireshark may not expose raw Wi-Fi headers when capturing through some wireless adapters.

Record the packet numbers and actual header values from your capture rather than inventing packet details.

2. Comparison of the OSI and TCP/IP Models

OSI Model (7 Layers)	**TCP/IP Model (4 Layers)**	**Real-World Example**
Application (7)	Application	Searching for a restaurant on Zomato
Presentation (6)	Application	Encrypting data sent through WhatsApp
Session (5)	Application	Maintaining a communication session with a service
Transport (4)	Transport	TCP delivering data reliably or UDP carrying real-time traffic

Network (3)	Internet	IP routing data toward Instagram's server
Data Link (2)	Network Access	Ethernet frames sent through a local switch
Physical (1)	Network Access	Electrical signals travelling through an Ethernet cable

The OSI model has seven layers, while the TCP/IP model groups related functions into four layers. The OSI model is useful for learning and troubleshooting; TCP/IP describes the protocol architecture used by the Internet.

3. TCP Segment Header: Fields and Functions

Use this guide to label the TCP header screenshot provided by your trainer.

TCP Header Fields

Source Port (16 bits)

Destination Port (16 bits)

Sequence Number (32 bits)

Acknowledgment Number (32 bits)

Header Length + Flags + Window Size

Checksum + Urgent Pointer

Simplified field layout; options and padding may also be present.

Field	**Function**
Source Port	Identifies the sending application's communication endpoint.
Destination Port	Identifies the receiving application's communication endpoint.
Sequence Number	Tracks the position of data bytes so TCP can put data in order.
Acknowledgment Number	Indicates the next byte expected from the other endpoint when acknowledgment is used.
Header Length (Data Offset)	Indicates where the TCP payload begins.
Flags	Control connection behavior, such as SYN, ACK, FIN and RST.
Window Size	Advertises how much data the receiver can accept.
Checksum	Detects corruption in the TCP header and data.
Urgent Pointer	Indicates urgent data boundaries when the URG flag is set.

Options	Carries additional settings, such as maximum segment size (MSS) and window scaling.

4. Diagram: Message Travels Through TCP/IP Layers

Example: Sending a message from my phone to Instagram.

1. Application Layer

I type and send a message in the app.

2. Transport Layer

TCP or another transport protocol carries application data; TCP can segment it and provide reliable delivery.

3. Internet Layer

IP addressing and routing move packets toward the destination.

4. Network Access Layer

Wi-Fi or Ethernet frames carry data over the local link as radio or electrical signals.

At the destination, the layers process the received data in reverse order.

Note: Instagram's exact transport protocol depends on the connection and service. Do not assume every Instagram message uses TCP.

5. Identify One Packet from Your Wireshark Capture

Use a packet that you actually captured. For example, a DNS query generated while loading Flipkart is often a useful packet to inspect, although DNS may be cached or encrypted.

Example write-up (if your capture contains a DNS query):

- Packet type: DNS query
- TCP/IP layer: Application
- How I identified it: Wireshark labels the packet as DNS and displays fields such as the queried domain name and query type.
- Real-world action: My browser needed to resolve a website's domain name to an IP address before connecting to the server.

You can also inspect the same packet at the other layers:

- Transport: UDP or TCP header, including port numbers.
- Internet: IP header with source and destination addresses.
- Network Access: Ethernet or Wi-Fi link-layer header, if available in the capture.